

Why Has the Price of Gold Risen So Rapidly Since 2022? Temporary Risk Premium or Persistent Structural Shift?

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Abstract

Since 2022, gold prices have risen sharply even as real interest rates have remained high, challenging the traditional view that higher real yields should reduce the appeal of holding a non-interest-bearing asset. This paper asks whether this rise reflects a temporary safe-haven premium or a more persistent change in gold-pricing dynamics. Using monthly data from 2003 to 2026, we examine the relationship between real gold returns and U.S. Treasury Inflation-Protected Securities (TIPS) yields, while controlling for trade policy uncertainty, geopolitical risk, financial market volatility, and financial stress. We also compare three measures of gold's real value: the CPI-adjusted gold price, the gold-to-oil ratio, and the gold-to-silver ratio. We find that before 2022, increases in real interest rates were strongly associated with lower real gold returns, consistent with the opportunity-cost channel. After the outbreak of the Russo-Ukrainian War, this relationship becomes much weaker and is no longer statistically significant in our main post-2022 specifications. Rolling-window estimates show a similar decline in gold's sensitivity to real rates, though the evidence remains time-varying and should be interpreted cautiously. Asymmetry tests further suggest that some uncertainty shocks, especially policy uncertainty, may have uneven effects on selected measures of gold's real value. Overall, the recent rise in gold prices is difficult to explain through real rates alone and may reflect a more durable reassessment of gold's safe-haven role, although the short post-2022 sample prevents a definitive conclusion about a permanent structural break.

Keywords: Gold prices, real interest rates, safe-haven demand, geopolitical risk, trade policy uncertainty

1 Introduction

Gold has long been viewed as a safe-haven asset, especially during periods of uncertainty, financial stress, or declining confidence in conventional financial assets. Because gold does not pay interest or dividends, the standard opportunity-cost framework predicts that higher real interest rates should reduce the relative attractiveness of holding gold. Since 2022, however, gold prices have risen rapidly even as real yields, measured by U.S. Treasury Inflation-Protected Securities (TIPS), have remained relatively high. This pattern raises the central question of this paper: has the traditional real-rate channel merely been temporarily overwhelmed by risk-related demand, or has the relationship between gold and real rates changed more persistently?

Recent media and policy discussions attribute the post-2022 gold surge to several forces, including trade-policy uncertainty, geopolitical tensions, and central-bank accumulation of gold reserves. We use this observation as motivation rather than as a substitute for empirical testing. In the current empirical specification, we directly examine trade-policy uncertainty, geopolitical risk, financial-market volatility, and financial stress. The central-bank reserve channel is not directly estimated in the current version because we have not yet incorporated a consistent monthly measure of central-bank gold purchases or reserves into the regression framework.

This capstone paper therefore has three objectives. First, it develops an asset-pricing indifference condition for gold that clarifies the role of real interest rates as an opportunity cost. Second, it tests whether the sensitivity of real gold returns to changes in real yields weakened after February 2022, the start of the full-scale Russian invasion of Ukraine. Third, it examines whether uncertainty shocks have asymmetric effects that may help explain why gold prices remain elevated even after some risk indicators decline. Throughout the paper, we interpret the evidence cautiously: the post-2022 sample is short, and the results are better viewed as suggestive evidence of persistent attenuation in the traditional real-rate channel than as definitive proof of a permanent structural break.

2 Literature Review

The paper builds on three strands of literature. The first examines whether gold functions as a hedge or safe haven. Baur and Lucey [3] distinguish a hedge from a safe haven and show that gold can behave as a safe haven during extreme stock-market conditions. Baur and McDermott [4] extend this question to international markets and find that gold's safe-haven role varies across countries and market conditions. This literature motivates our treatment of gold as an asset whose return may be affected by risk and uncertainty in addition to conventional macroeconomic fundamentals.

The second strand studies the relationship between gold, inflation, and real interest rates. Barsky and Summers [2] connect real interest rates and the gold standard in the context of Gibson's paradox, while Erb and Harvey [7] emphasize that popular narratives about gold, including its role as an inflation hedge, should be evaluated carefully against long-run evidence. This literature motivates our focus on real rather than nominal gold prices and on TIPS yields as measures of the real opportunity cost of holding gold.

The third strand concerns uncertainty measurement and time-series econometrics. We use the Geopolitical Risk Index of Caldara and Iacoviello [5] and the Economic Policy Uncertainty framework of Baker, Bloom, and Davis [1] as observable proxies for risk-related demand. Our empirical design also relies on standard time-series tools, including Augmented Dickey-Fuller unit-root testing [6] and Newey-West heteroskedasticity-and-autocorrelation-consistent standard errors [8]. These methods help ensure that the regression analysis is not driven by spurious correlation or by misleading inference under serial correlation.

3 Theoretical Framework

This section develops an asset-pricing indifference condition for gold. The condition is conceptually similar to the logic behind uncovered interest parity, but it is not itself a UIP condition. UIP applies an asset-pricing no-arbitrage idea to the choice between two currency-denominated assets. Here, the comparison is between holding gold and holding a safe dollar asset. The purpose of the framework is therefore to identify the expected return on gold that would make an investor indifferent, before future information is revealed, between the two investment strategies.

3.1 Nominal Asset-Pricing Indifference Condition

Suppose a rational investor chooses at time t between holding a safe dollar asset and holding one ounce of gold. Let R_t denote the nominal interest rate on the safe dollar asset, G_t the current dollar price of one ounce of gold, and $G_{t+1}^e = E_t(G_{t+1})$ the expected dollar price of gold one period ahead.¹

If the investor puts G_t dollars into the safe asset, the expected dollar value next period is $G_t(1 + R_t)$. If instead the investor buys one ounce of gold, the expected dollar value next period is G_{t+1}^e . The indifference condition is therefore

$$G_t(1 + R_t) = G_{t+1}^e. \quad (1)$$

Equivalently, in expected-return form,

$$R_t = \frac{G_{t+1}^e - G_t}{G_t}. \quad (2)$$

Equation (2) is an ex-ante asset-pricing condition: it compares returns using the information available at time t . It does not state that the realized return on gold must equal the safe return ex post.

3.2 Real Asset-Pricing Indifference Condition

The nominal condition is not sufficient for our empirical question because the dollar price of gold is affected by inflation. Using the Fisher approximation, the nominal interest rate R_t can be decomposed into the real interest rate r_t and expected inflation π_{t+1}^e :

$$R_t \approx r_t + \pi_{t+1}^e. \quad (3)$$

¹The superscript e denotes an expectation formed at time t . Throughout Section 3, a single unit of time is defined as one year for exposition. In the empirical section, the data are monthly.

Higher expected inflation may increase the nominal price of gold, while higher real rates raise the opportunity cost of holding a non-interest-bearing asset. To isolate the opportunity-cost channel, we work with real measures of gold's purchasing power. Let g_t denote the real gold price. We use three empirical measures of g_t .

1. CPI-Adjusted Gold Price ($g_{CPI,t}$). This measure deflates the nominal gold price using the U.S. Consumer Price Index (CPI). It represents gold's purchasing power relative to a representative U.S. consumption basket. This approach is intuitive for measuring gold's ability to preserve purchasing power, but it implicitly treats U.S. consumer prices as the relevant deflator for a globally traded asset.

2. Gold-to-Crude Oil Ratio ($g_{Oil,t}$). This ratio divides the nominal gold price by the nominal crude-oil price. It measures how many barrels of crude oil can be purchased with one ounce of gold. Because both assets are denominated in U.S. dollars, the ratio reduces the influence of dollar inflation. Its disadvantage is that it can be heavily affected by oil-specific shocks, especially during energy crises or geopolitical events.

3. Gold-to-Silver Ratio ($g_{Silver,t}$). This ratio divides the nominal gold price by the nominal silver price. Since both assets are precious metals, the ratio helps isolate gold's relative monetary and financial value. However, because silver also has safe-haven properties and substantial industrial demand, this measure may partially remove the uncertainty premium that is specific to gold.

Expressed in real terms, the basic indifference condition becomes

$$r_t = \frac{g_{t+1}^e - g_t}{g_t}. \quad (4)$$

To avoid confusion with interest-parity terminology, we interpret Equation (4) as a real asset-pricing indifference condition rather than as a literal "real interest parity" condition.

3.3 Risk and Safe-Haven Premium

Equation (4) abstracts from the fact that investors may value gold for reasons not captured by its expected monetary return alone. Gold may provide insurance-like benefits during episodes of financial stress, geopolitical risk, or declining confidence in fiat currencies.

We represent these benefits through a time-varying risk or safe-haven premium, ρ_t :

$$\underbrace{r_t}_{\text{real opportunity cost}} + \underbrace{\rho_t}_{\text{risk/safe-haven premium}} = \underbrace{\frac{g_{t+1}^e - g_t}{g_t}}_{\text{expected real return on gold}}. \quad (5)$$

The sign of ρ_t is not imposed by definition. Ex ante, it captures the wedge between the safe real return and the expected real return on gold that is consistent with investor indifference. In some periods, an insurance or convenience-yield motive may lead investors to accept a lower expected monetary return on gold; in that case, the inferred premium can be negative. In other periods, gold-specific risk may require compensation, and the premium can be positive. The sign and magnitude of ρ_t are therefore empirical questions rather than assumptions built into the equation.

It is important to separate the ex-ante condition from the ex-post realized return. At time t , investors choose between gold and safe assets before g_{t+1} is observed. Let the realized real return on gold be

$$\Delta g_{t+1} = \frac{g_{t+1} - g_t}{g_t}. \quad (6)$$

Under rational expectations, the realized return equals the expected return plus a forecast error:

$$\Delta g_{t+1} = \frac{g_{t+1}^e - g_t}{g_t} + \epsilon_{t+1}, \quad E_t(\epsilon_{t+1}) = 0. \quad (7)$$

Substituting Equation (5) into Equation (7) gives

$$\Delta g_{t+1} = r_t + \rho_t + \epsilon_{t+1}. \quad (8)$$

Because ρ_t is unobservable, the empirical analysis uses observable proxies for uncertainty and financial stress. The reduced-form empirical relationship is

$$\Delta g_t = \alpha + \beta r_t + \mathbf{X}_t \gamma + \varepsilon_t, \quad (9)$$

where $\mathbf{X}_t$ contains observable proxies for the risk and safe-haven premium. The coefficient β measures the sensitivity of gold returns to the real interest-rate channel after controlling for these proxies. A decline in the magnitude or significance of β after 2022 would be consistent with attenuation of the traditional opportunity-cost channel, but it would not by itself prove a permanent structural break.

4 Data

4.1 Sample Period and Frequency

We use monthly frequency data spanning from January 2003 to February 2026. While daily data introduces a significant amount of noise, quarterly or yearly data are insufficient for robust statistical inference. Thus, the use of monthly data allows us to strike a balance between data quality and the number of observations.

4.2 Gold Prices and Real Returns

The primary dependent variable in our analysis is the real return on gold. We obtain the nominal gold price, nominal crude oil price, nominal silver price, and the CPI-adjusted gold price ($g_{CPI,t}$) directly from the Macrotrends website. Then, following the measures defined in Section 3.2, we construct the Gold-to-Oil Ratio ($g_{Oil,t}$) and the Gold-to-Silver Ratio ($g_{Silver,t}$) by dividing the nominal gold price by the nominal prices of crude oil and silver, respectively, to identify gold's true purchasing power.

4.3 Real Interest Rates

To represent the real opportunity cost of holding gold, which is our core independent variable, we primarily use the yield on 10-year U.S. Treasury Inflation-Protected Securities (TIPS). Specifically, we use the "Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity, Quoted on an Investment Basis, Inflation-Indexed." For robustness checks, we also incorporate 5-year TIPS yield rates. Both series are sourced from the Federal Reserve Economic Data (FRED) database.

4.4 Safe-Haven Premium Proxies

Several explanatory variables are used as proxies for the safe-haven premium (ρ_t) that controls for market sentiment and geopolitical shocks:

1. Gold Monthly Realized Volatility. This variable captures the intrinsic risk and price uncertainty within the gold market. Following the standard definition in financial econometrics, we construct the monthly realized volatility (RV) by using daily gold spot price (G_d) and calculating the square root of the sum of squared daily logarithmic returns (y_d) within each month:

$$y_d = \ln(G_d) - \ln(G_{d-1}) \quad (10)$$

The monthly realized volatility for month t is then defined as:

$$RV_t = \sqrt{\sum_{d=1}^N y_{d,t}^2} \times 100\% \quad (11)$$

where N is the number of trading days in month t . In our computation, the return for the first trading day of each month is calculated using the last trading day of the preceding month and is included in the current month's summation. Multiplying by 100% expresses the volatility in percentage points.

Due to data availability, for the period before 2017, we utilize historical daily gold fixing prices from the London Bullion Market, sourced via The Financial Forecast Center. Since 2017, daily price data are obtained directly from Macrotrends to maintain consistency with our primary monthly data source.

2. Geopolitical Risk. We use the Geopolitical Risk Index (GPR) developed by Caldara and Iacoviello (2022) as a proxy for safe-haven demand. The index is constructed via automated text searches across ten leading newspapers and tallies articles concerning geopolitical tensions across eight categories. High GPR levels historically signal increased global downside risks and may therefore increase safe-haven demand for gold.

3. Trade Policy Uncertainty. We use the Economic Policy Uncertainty Index for Trade Policy², obtained from the database developed by Baker, Bloom, and Davis. This index reflects the frequency of news coverage regarding trade policy shifts, assuming that stronger media coverage indicates a more uncertain trade environment.

4. Financial Market Volatility. Often referred to as the “fear index,” the Volatility Index (VIX)³ is published by the CBOE and sourced from FRED. It reflects expected stock market volatility derived from option prices; a rise in the VIX generally suggests that investors anticipate greater market risk.

5. Financial Stress Index. We use the St. Louis Fed Financial Stress Index (FSI)⁴, which combines various financial variables, such as interest rates and yield spreads, to capture

²Source: Federal Reserve Bank of St. Louis, FRED, “Economic Policy Uncertainty Index: Categorical Index: Trade Policy,” <https://fred.stlouisfed.org/series/EPUTRADE>.

³Source: Federal Reserve Bank of St. Louis, FRED, “CBOE Volatility Index: VIX,” <https://fred.stlouisfed.org/series/VIXCLS>.

⁴Source: Federal Reserve Bank of St. Louis, FRED, “St. Louis Fed Financial Stress Index,” <https://fred.stlouisfed.org/series/STLFSI4>.

overall systemic stress. A higher index value indicates more strained financial market conditions than usual.

5 Empirical Methodology

In this section, we outline the econometric methods used to investigate the relationship between gold prices and real interest rates. Our empirical strategy moves from basic stationarity tests to dynamic and non-linear models to capture the breakdown of the traditional pricing relationship.

5.1 Unit Root Testing

Before estimating the regression models, we examine the time-series properties of all variables used in the analysis. This step is important because regressions involving non-stationary variables may produce spurious relationships. A variable is classified as $I(0)$ if it is stationary in levels. It is classified as $I(1)$ if its first difference is stationary, and $I(2)$ if its second difference is stationary.

We use the Augmented Dickey-Fuller (ADF) test to test for unit roots. For a generic variable y_t , the ADF regression can be written as

$$\Delta y_t = \alpha + \delta t + \phi y_{t-1} + \sum_{j=1}^p \theta_j \Delta y_{t-j} + u_t, \quad (12)$$

where the null hypothesis is

$$H_0 : \phi = 0, \quad (13)$$

which means that the series contains a unit root. Rejection of the null provides evidence that the series is stationary under the chosen deterministic specification.

Because the correct deterministic component is not always clear in advance, we run the ADF test under three alternative model specifications:

- (i) **Trend specification:** the ADF regression includes both an intercept and a linear time trend.
- (ii) **Constant specification:** the ADF regression includes an intercept but no linear time trend.
- (iii) **None specification:** the ADF regression includes neither an intercept nor a linear time trend.

These three specifications are not treated as a required testing sequence. Instead, they represent alternative assumptions about the deterministic component of the data-generating process. Reporting them side by side allows us to check whether the stationarity conclusion is sensitive to the inclusion of an intercept or trend.

For each variable, we conduct the ADF tests at three differencing orders: the level series, the first difference, and the second difference. In Stata notation, these correspond to y_t , Δy_t , and $\Delta^2 y_t$. Since the data are monthly, we use 12 lags in the ADF regression to allow for annual serial correlation. Thus, the lag length is fixed at $p = 12$ across all reported tests.

The classification rule is based on the lowest differencing order at which the unit-root null is rejected at the 5% level. Specifically, if the null is rejected for the level series under at least one of the three deterministic specifications, the variable is classified as $I(0)$. If the level series fails to reject the null but the first difference rejects it, the variable is classified as $I(1)$. If rejection occurs only after taking the second difference, the variable is classified as $I(2)$.

5.2 Baseline Segmented Regression

To test whether the relationship between gold returns and real yields has weakened, we estimate a baseline segmented regression. Based on the reduced-form relationship in Equation (9), we use changes in real interest rates and observable proxies for the risk/safe-haven premium.⁵ Throughout the empirical analysis, we reserve Δ for the monthly first-difference operator and denote the 12-month year-on-year real return on gold as $\tilde{g}_t$:

$$\Delta \tilde{g}_t = \alpha + \beta \Delta r_t + \mathbf{S}_t \gamma + \varepsilon_t. \quad (14)$$

In this specification, $\tilde{g}_t$ represents the year-on-year (YoY) real return on gold:

$$\tilde{g}_t = \frac{g_t - g_{t-12}}{g_{t-12}}, \quad (15)$$

where g_t is the real gold price in month t . Since $\tilde{g}_t$ is classified as $I(1)$ in Table 1, we use its first difference $\Delta \tilde{g}_t$ as the dependent variable. The core independent variable Δr_t is the first difference of the TIPS yield, representing the monthly change in the real opportunity cost of holding gold. The vector $\mathbf{S}_t$ includes gold monthly realized volatility (RV), the change in the Economic Policy Uncertainty index (ΔEPU), the CBOE Volatility Index (VIX), the St. Louis Fed Financial Stress Index (FSI), and the change in the Geopolitical Risk index (ΔGPR). We estimate the model using Newey-West standard errors with 4 lags to allow for heteroskedasticity and autocorrelation.

We split the sample at February 2022, the start month of the full-scale Russian invasion of Ukraine. This split is motivated by the sharp increase in geopolitical uncertainty and the unusual post-2022 combination of rising real yields and rising gold prices. The

⁵Different from Section 3, t denotes the current month in the empirical analysis.

split-sample design allows us to compare the real-rate coefficient β before and after the shock. Because the post-2022 sample contains only 48 monthly observations, the post-war estimates should be interpreted as suggestive evidence rather than as definitive proof of a permanent structural break.

5.3 Rolling Window Analysis

The segmented regression compares two broad periods but does not show how the real-rate sensitivity evolves over time. To examine this evolution, we implement a 48-month rolling-window regression. We re-estimate the baseline model over a moving 48-month window and track the estimated coefficient $\hat{\beta}$. This exercise is intended as a descriptive stability check rather than a formal structural-break test. It helps assess whether the weakening of the opportunity-cost channel after 2022 is unusual relative to earlier episodes, such as the Global Financial Crisis and the Covid-19 pandemic.

5.4 Asymmetric Shock Response

Finally, we investigate whether gold prices respond differently to bad news compared to good news. Investors might rush into gold during crises but be slower to exit when risks subside. To test this, we decompose the changes in each safe-haven proxy into positive and negative components. Take the GPR index as an example:

$$\begin{aligned}\Delta GPR_t^+ &= \begin{cases} \Delta GPR_t, & \text{if } \Delta GPR_t > 0 \\ 0, & \text{if } \Delta GPR_t \leq 0 \end{cases} \\ \Delta GPR_t^- &= \begin{cases} \Delta GPR_t, & \text{if } \Delta GPR_t < 0 \\ 0, & \text{if } \Delta GPR_t \geq 0 \end{cases}\end{aligned}\tag{16}$$

where $\Delta GPR_t = GPR_t - GPR_{t-1}$. Here, GPR_t^+ represents an increase in geopolitical risk, while GPR_t^- represents a decrease in geopolitical risk. We then estimate the following regression:

$$\Delta \tilde{g}_t = \alpha + \beta \Delta r_t + \gamma^+ \Delta GPR_t^+ + \gamma^- \Delta GPR_t^- + \mathbf{S}'_t \boldsymbol{\gamma}' + u_t\tag{17}$$

where $\mathbf{S}'_t$ includes other control variables. We perform a Wald test to evaluate the null hypothesis of symmetry:

$$\begin{aligned}H_0 : \gamma^+ &= \gamma^- \\ H_1 : \gamma^+ &> \gamma^-\end{aligned}\tag{18}$$

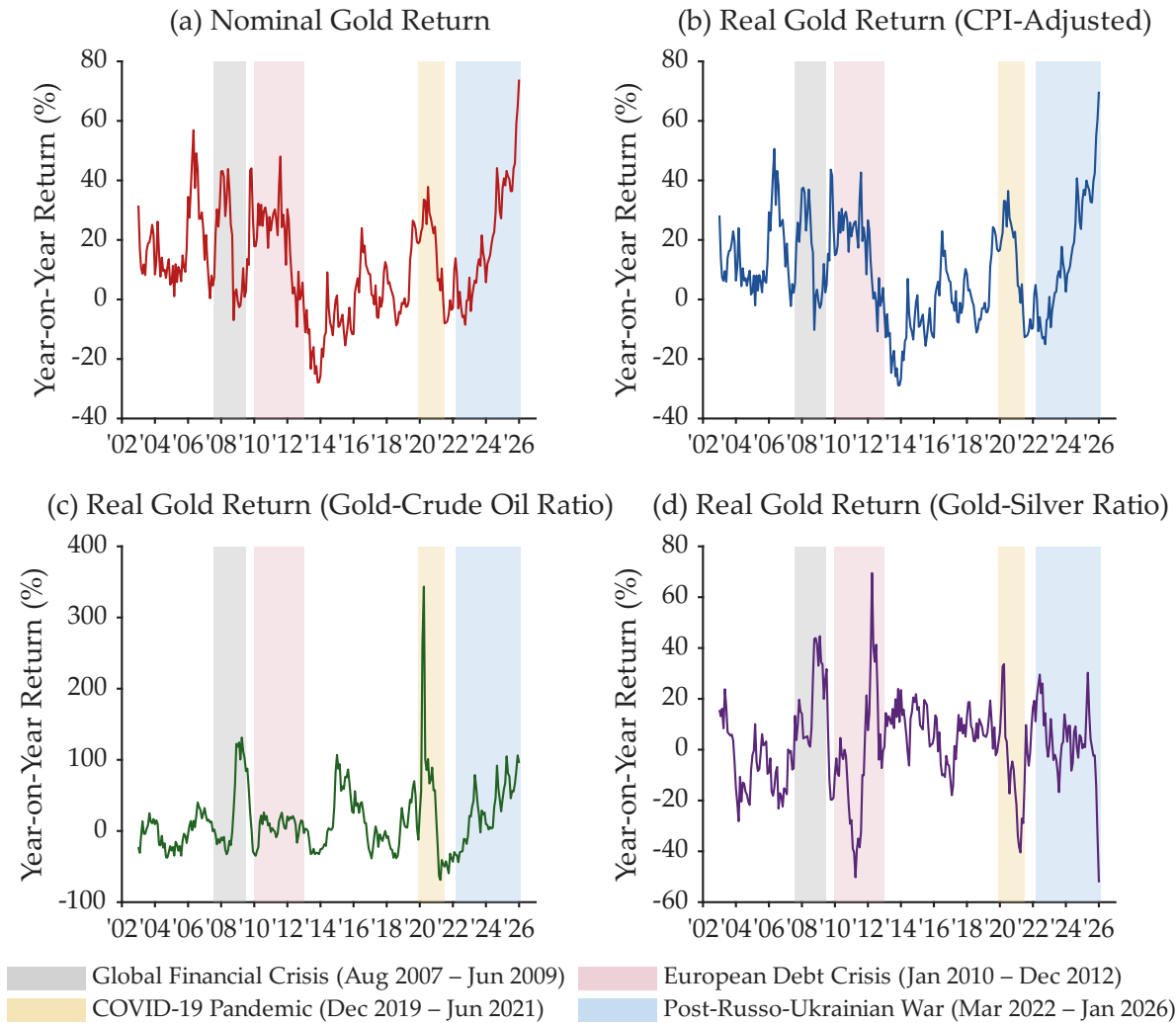
If γ^+ is significantly larger than γ^- , it would provide empirical evidence of a ratchet effect, explaining why gold prices remain high even as the initial shocks that drove them upward begin to fade. The same test is applied to other safe-haven proxies.

6 Results

6.1 Trend Charts

We first examine time-series plots of our key variables. Figure 1 illustrates the year-on-year gold returns.

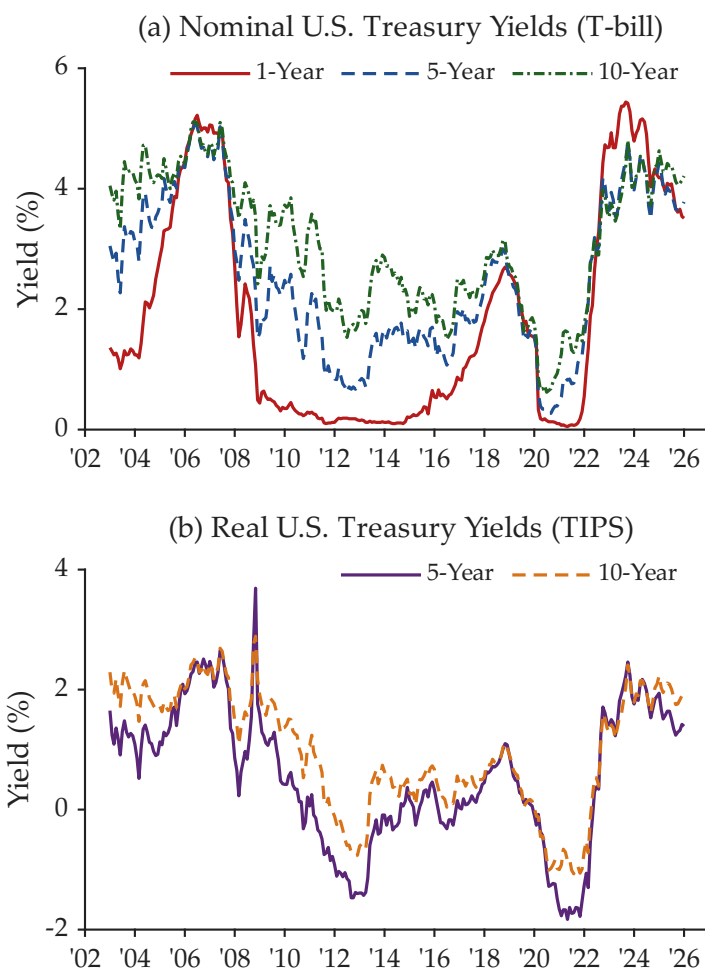
Figure 1: Year-on-Year Gold Returns with Different Deflators



Across the four panels in Figure 1, gold returns tend to rise during several major stress episodes, although the magnitude differs substantially across deflators. The nominal and CPI-adjusted measures show particularly strong post-2022 performance, while the gold-oil ratio is more volatile because it also reflects oil-market shocks. Figure 2 shows that nominal Treasury yields and real TIPS yields increased sharply after 2022. Under the

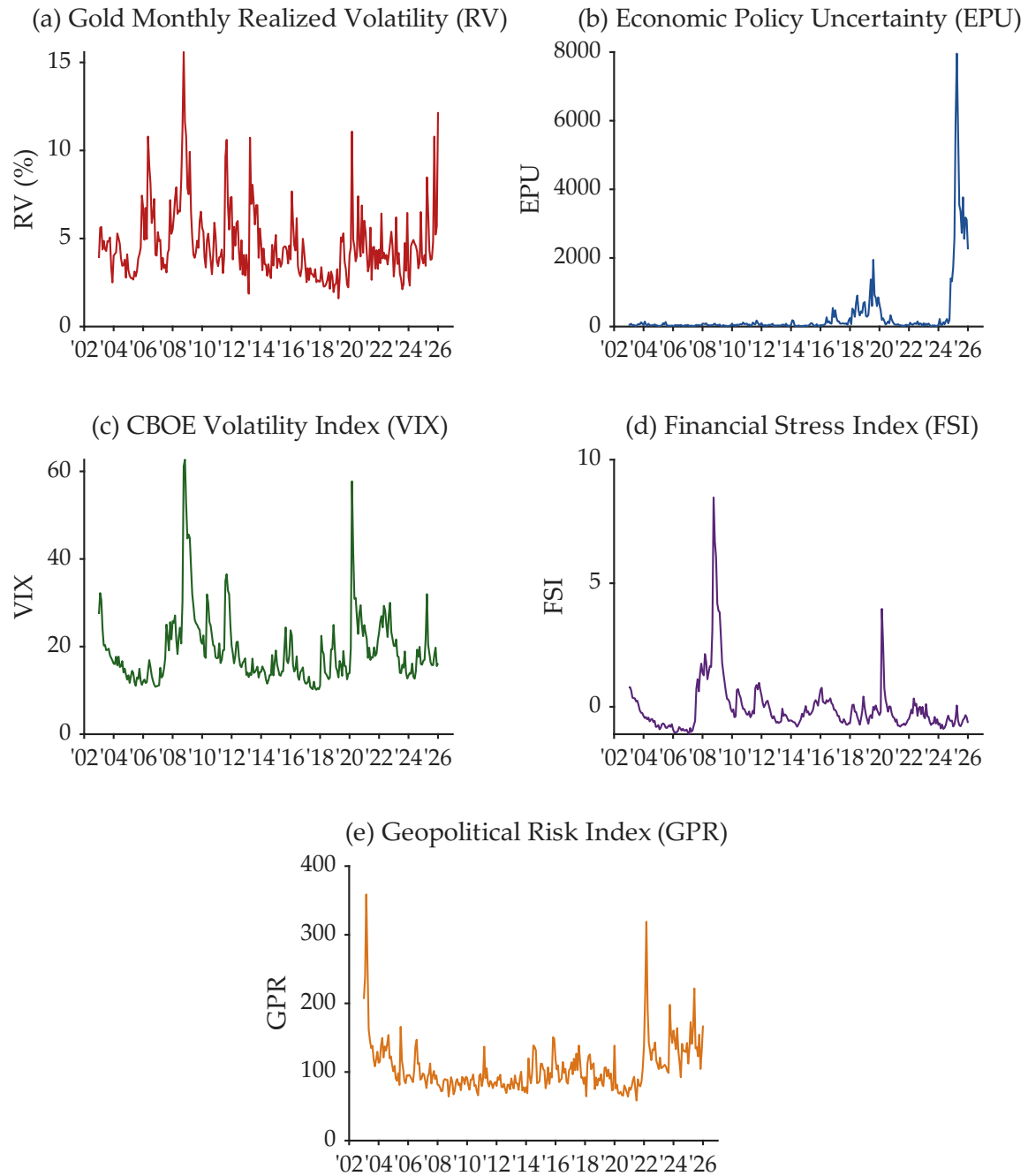
standard opportunity-cost channel, higher real yields should place downward pressure on gold returns. The simultaneous rise in real yields and gold prices therefore provides descriptive motivation for testing whether the real-rate channel weakened after 2022.

Figure 2: Nominal and Real U.S. Treasury Yields



Finally, Figure 3 plots the five observable proxies used to capture risk and safe-haven demand: gold realized volatility (RV), the Trade Policy Uncertainty Index (EPU), the CBOE Volatility Index (VIX), the St. Louis Fed Financial Stress Index (FSI), and the Geopolitical Risk Index (GPR). These series capture different forms of uncertainty and stress. Their movements are used as controls rather than as direct observations of the unobservable premium ρ_t .

Figure 3: Proxies for the Safe-Haven Premium



6.2 Unit Root Test Results

The Augmented Dickey-Fuller (ADF) results are summarized in Table 1. The table reports ADF t -statistics under alternative deterministic specifications and classifies each variable according to the lowest differencing order at which the unit-root null is rejected. The main point for the regression design is that several variables, including the CPI-adjusted year-on-year gold return, the 10-year TIPS yield, EPU, and GPR, are treated as $I(1)$ and are therefore used in first differences.

Table 1: ADF Unit Root Test Results

Variable	Level			First Diff.			Second Diff.
	Trend	Const.	None	Trend	Const.	None	Trend
<i>Panel A: Stationary in Levels — $I(0)$</i>							
YOY of Gold–Oil Ratio	−3.632**	—	—	—	—	—	—
YOY of Gold–Silver Ratio	−3.015	−3.112**	—	—	—	—	—
1-yr Treasury Yield	−3.030	−3.025**	—	—	—	—	—
5-yr TIPS Yield [†]	−2.201	−2.237	−1.987**	—	—	—	—
Gold Realized Volatility	−3.152*	−3.185**	—	—	—	—	—
VIX	−2.993	−2.997**	—	—	—	—	—
FSI	−3.175*	−3.049**	—	—	—	—	—
<i>Panel B: Integrated of Order One — $I(1)$</i>							
YOY Gold Return	−0.424	−0.635	0.108	−6.439***	—	—	—
CPI-Adj. YOY Gold Return	−0.463	−0.701	−0.087	−6.176***	—	—	—
5-yr Treasury Yield	−2.141	−2.204	−0.925	−3.516**	—	—	—
10-yr Treasury Yield	−1.539	−1.805	−0.607	−4.239***	—	—	—
10-yr TIPS Yield	−1.727	−1.920	−1.352	−4.082***	—	—	—
EPU	−0.836	−0.137	0.401	−5.108***	—	—	—
GPR	−3.060	−2.505	−0.497	−7.372***	—	—	—
Crude Oil	−2.836	−2.870*	−0.370	−5.270***	—	—	—
Gold–Oil Ratio	−1.722	0.257	1.179	−4.104***	—	—	—
Gold–Silver Ratio	−2.918	−1.863	−0.361	−5.175***	—	—	—
<i>Panel C: Integrated of Order Two — $I(2)$</i>							
Gold	1.764	2.170	1.106	1.947	2.479	2.807	−5.492***
CPI-adj. Gold	0.234	1.500	1.285	0.341	0.613	0.967	−6.453***
Silver	0.226	1.151	1.386	−0.939	−0.847	−0.619	−4.121***

Notes: The null hypothesis of the ADF test is that the series contains a unit root. ADF t -statistics are reported for three alternative deterministic specifications: intercept with trend (*Trend*), intercept only (*Const.*), and no deterministic terms (*None*). These specifications are reported side by side for transparency and should not be interpreted as a required testing sequence. For each variable, tests are conducted in levels, first differences, and second differences, using 12 lags in all ADF regressions. The reported integration order is based on the lowest differencing order at which the unit-root null is rejected at the 5% level under at least one deterministic specification. *, **, and *** denote rejection of the unit-root null at the 10%, 5%, and 1% significance levels, respectively. [†] The 5-year TIPS yield is classified as $I(0)$ only under the Level/None specification. Because a zero-mean specification is economically unusual for an interest-rate series, this classification is ambiguous. We therefore include both the level and first difference of the 5-year TIPS yield as robustness checks in the empirical analysis.

6.3 Baseline Regression Results

Table 2 presents our baseline Newey-West estimates across three dependent variables (Panels A–C), three real rate proxies (columns 1–3 within each sample group), and three sample periods (full sample, pre-February 2022, and post-February 2022).

The Opportunity Cost Channel. Panel A and Panel B provide evidence that the opportunity-cost channel was economically and statistically important before February 2022. In Panel A, where the dependent variable is the first difference of the CPI-adjusted year-on-year gold return, the pre-February 2022 coefficient on the 10-year TIPS change is -10.380 and significant at the 1% level, while the coefficient on the 5-year TIPS change is -5.254 and significant at the 5% level. Panel B, based on the gold-oil ratio, yields larger negative coefficients of -43.992 and -27.570 , both significant at the 5% level. These results are consistent with the asset-pricing indifference framework: increases in real yields raise the opportunity cost of holding gold and are associated with lower real gold returns.

The full-sample estimates also show negative and statistically significant coefficients on Δr^{10} and Δr^5 in Panels A and B. However, the full-sample estimates pool together periods in which the relationship may have changed. For this reason, the split-sample estimates are more informative for the post-2022 research question. The level specification r^5 should be interpreted cautiously because the stationarity classification of the 5-year TIPS level is ambiguous, as discussed in Table 1.

Attenuation after February 2022. The post-February 2022 estimates suggest a sharp weakening of the negative real-rate relationship. In Panels A and B, the coefficients on Δr^{10} and Δr^5 become statistically indistinguishable from zero. In Panel A, the coefficients fall in magnitude to -2.166 and -2.541 ; in Panel B, the corresponding estimates are -10.044 and -16.446 , but they are imprecisely estimated. These results are consistent with a post-2022 attenuation of the traditional opportunity-cost channel.

This finding should not be overstated. The post-war sample contains only 48 monthly observations, and the level specification r^5 remains positive and marginally significant in some columns. Therefore, the evidence does not prove that gold has permanently decoupled from real rates. A more cautious interpretation is that, after February 2022, real-rate changes no longer explain gold returns as strongly as they did in the pre-war period, after controlling for the available safe-haven proxies.

Panel C: The Gold-Silver Ratio as an Outlier. Panel C presents a different pattern. The real-rate coefficient is statistically insignificant across all three sample periods and

Table 2: Baseline Segmented Regression Results

	Full Sample			Pre-Feb 2022			Post-Feb 2022		
Real Rate:	(1) Δr^{10}	(2) Δr^5	(3) r^5	(4) Δr^{10}	(5) Δr^5	(6) r^5	(7) Δr^{10}	(8) Δr^5	(9) r^5
<i>Panel A: Dependent Variable = $\Delta \tilde{g}_{CPI,t}$ (First Difference of CPI-Adjusted YoY Gold Return)</i>									
Real Rate	−8.573***	−5.081***	0.680**	−10.380***	−5.254**	0.553	−2.166	−2.541	1.200**
RV	−0.125	−0.130	−0.238	−0.421	−0.466	−0.626*	0.696*	0.677*	0.752**
Δ EPU	0.001	0.001	0.001	0.002*	0.003**	0.003**	0.001	0.001	0.001
VIX	0.013	0.016	0.060	−0.051	−0.016	0.078	−0.199	−0.186	−0.189
FSI	−0.388	−0.372	−0.573	0.420	0.278	−0.205	1.275	1.371	2.375
Δ GPR	0.006	0.006	0.010	0.003	0.002	0.003	0.015	0.014	0.015
Observations	277	277	277	229	229	229	48	48	48
R^2	0.053	0.037	0.019	0.073	0.048	0.027	0.111	0.115	0.119
Adjusted R^2	0.032	0.016	−0.002	0.048	0.023	0.001	−0.019	−0.015	−0.010
RMSE	7.263	7.324	7.391	7.462	7.560	7.643	5.876	5.864	5.850
<i>Panel B: Dependent Variable = $\Delta g_{Oil,t}$ (Gold-Oil Ratio Return)</i>									
Real Rate	−47.620***	−35.440***	6.077	−43.991**	−27.570**	2.065	−10.044	−16.446	17.223*
RV	1.517	1.565	0.711	−1.763	−1.830	−2.614	8.249***	8.048***	8.647***
Δ EPU	−0.001	−0.001	−0.001	−0.014	−0.011	−0.011	0.008	0.007	0.006
VIX	2.378	2.378	2.751*	2.721	2.819	3.248*	−1.583	−1.381	−0.828
FSI	−0.633	−0.555	−2.374	2.883	2.459	0.394	−27.183	−27.273	−15.159
Δ GPR	−0.030	−0.034	−0.006	−0.076	−0.081	−0.079	0.094	0.080	0.073
Observations	277	277	277	229	229	229	48	48	48
R^2	0.222	0.222	0.209	0.301	0.296	0.279	0.329	0.334	0.380
Adjusted R^2	0.205	0.205	0.192	0.282	0.277	0.260	0.230	0.236	0.290
RMSE	42.048	42.039	42.395	40.477	40.613	41.096	35.602	35.462	34.202
<i>Panel C: Dependent Variable = $\Delta g_{Silver,t}$ (Gold-Silver Ratio Return)</i>									
Real Rate	6.502	6.804	−0.470	2.193	3.995	−1.604	−12.453	−7.907	−1.380
RV	−1.736**	−1.763**	−1.654*	−1.170	−1.228*	−1.025	−3.780**	−3.727**	−3.552**
Δ EPU	−0.002	−0.001	−0.002	0.001	0.001	0.001	−0.005**	−0.005**	−0.004**
VIX	−0.123	−0.119	−0.157	−0.598	−0.577	−0.735*	2.397**	2.274**	1.953*
FSI	9.520***	9.513***	9.646***	11.979***	11.914***	12.943***	−8.027	−6.706	−6.618
Δ GPR	−0.001	0.000	−0.004	0.009	0.009	0.007	−0.016	−0.016	−0.004
Observations	277	277	277	229	229	229	48	48	48
R^2	0.236	0.240	0.232	0.263	0.265	0.271	0.478	0.467	0.458
Adjusted R^2	0.219	0.224	0.215	0.243	0.245	0.252	0.402	0.389	0.378
RMSE	15.455	15.409	15.492	15.554	15.528	15.463	12.127	12.255	12.361

Notes: All estimates use Newey-West standard errors with 4 lags. The sample spans January 2003 to February 2026 at monthly frequency. The pre/post split follows the onset of the 2022 Russian invasion of Ukraine (February 2022), yielding 229 pre-war and 48 post-war observations. Real rate proxies are: Δr^{10} = first difference of 10-year TIPS yield; Δr^5 = first difference of 5-year TIPS yield; r^5 = level of 5-year TIPS yield, included as a robustness check because its unit-root classification is ambiguous; see Table 1. Control variables include gold monthly realized volatility (RV), first difference of the Economic Policy Uncertainty index (Δ EPU), CBOE Volatility Index (VIX), St. Louis Fed Financial Stress Index (FSI), and first difference of the Geopolitical Risk index (Δ GPR). R^2 , adjusted R^2 , and RMSE are reported as model-fit measures for each specification. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

all three rate proxies, suggesting that the gold-silver ratio is not well explained by the opportunity-cost channel in this specification. This is consistent with the discussion in Section 3.2: because silver itself has both safe-haven and industrial-demand components, deflating gold by silver may remove part of the uncertainty premium that distinguishes gold from other assets. The Financial Stress Index (FSI) is positive and highly significant in the full sample and pre-war period, suggesting that systemic financial stress may affect gold and silver differently.

Safe-Haven Proxies. The safe-haven control variables are not uniformly significant across panels. Realized volatility (RV) is positive and significant in the post-war CPI-adjusted specification and strongly significant in the post-war gold-oil specification, suggesting that gold-specific volatility is associated with higher gold returns during the post-2022 period. EPU is positive and significant in the pre-war CPI-adjusted specifications but not consistently significant elsewhere. VIX, FSI, and GPR do not display stable significance in Panels A and B. This mixed evidence suggests that the observable proxies capture only part of the unobservable risk/safe-haven premium ρ_t .

Summary. Taken together, Table 2 supports a cautious version of the main hypothesis. The real-rate channel is clearly visible in the pre-February 2022 period for the CPI-adjusted and gold-oil measures, but it weakens substantially after February 2022. This pattern is consistent with a more important role for time-varying safe-haven demand in the post-2022 gold market. However, because the post-war sample is short and some coefficients are imprecise, the table should be interpreted as evidence of attenuation rather than conclusive evidence of a permanent structural break.

6.4 Rolling Window Evidence

To examine whether the sensitivity of gold returns to real interest rates changes over time, we estimate a 48-month rolling-window version of the baseline Newey-West regression. At each ending month from January 2007 to February 2026, we re-estimate the following specification over the most recent 48 monthly observations:

$$\Delta \tilde{g}_{CPI,t} = \alpha + \beta \Delta r_t^{10} + \gamma_1 RV_t + \gamma_2 \Delta EPU_t + \gamma_3 VIX_t + \gamma_4 FSI_t + \gamma_5 \Delta GPR_t + \mu_t. \quad (19)$$

The coefficient of interest is $\hat{\beta}$, which measures the time-varying sensitivity of the first-differenced CPI-adjusted year-on-year gold return to changes in the 10-year TIPS yield. Standard errors are Newey-West corrected with four lags in each rolling window. Figure 4 reports the rolling estimates together with a conventional 95% confidence band, while Figure 5 reports the same estimates with a one-standard-error band. The vertical red band marks February 2022, the onset of the Russo-Ukrainian War.

Figure 4: 48-Month Rolling Window Estimate of $\hat{\beta}$ (Sensitivity of $\Delta \tilde{g}_{CPI,t}$ to Δr_t^{10})

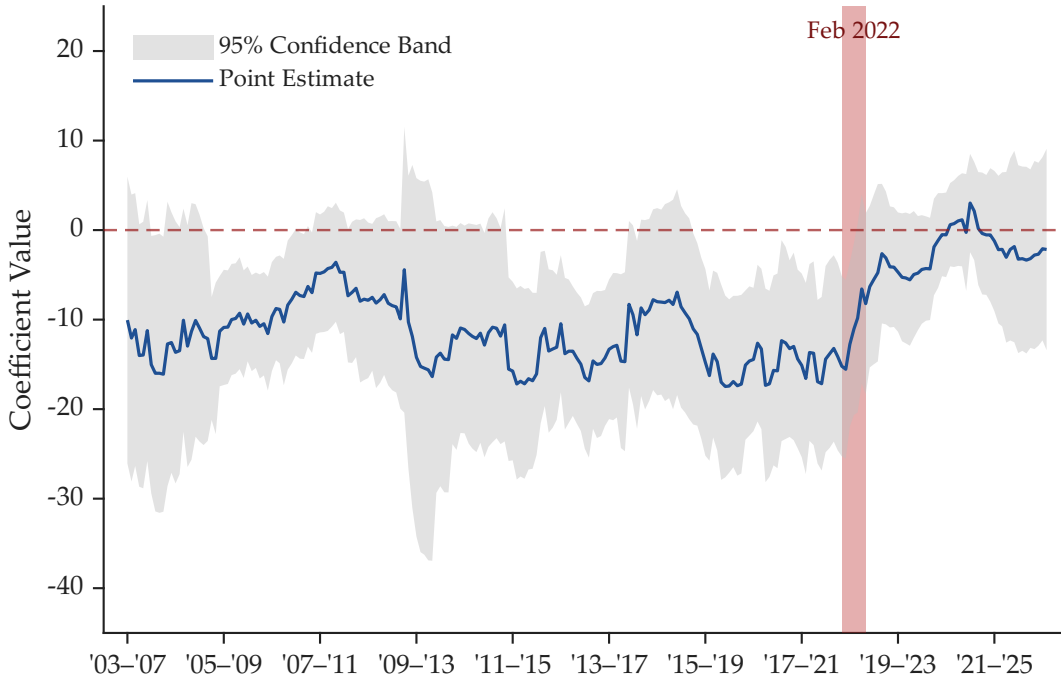


Figure 4 shows that $\hat{\beta}$ is negative for most of the pre-2022 period. This supports the opportunity-cost channel: when real interest rates rise, gold becomes less attractive

because it does not pay interest. At the same time, the 95% confidence band is fairly wide and includes zero in some windows, especially around crisis periods. This means that the pre-2022 relationship was not perfectly stable, even though the point estimates were mostly negative.

Figure 5: 48-Month Rolling Window Estimate with One-Standard-Error Band

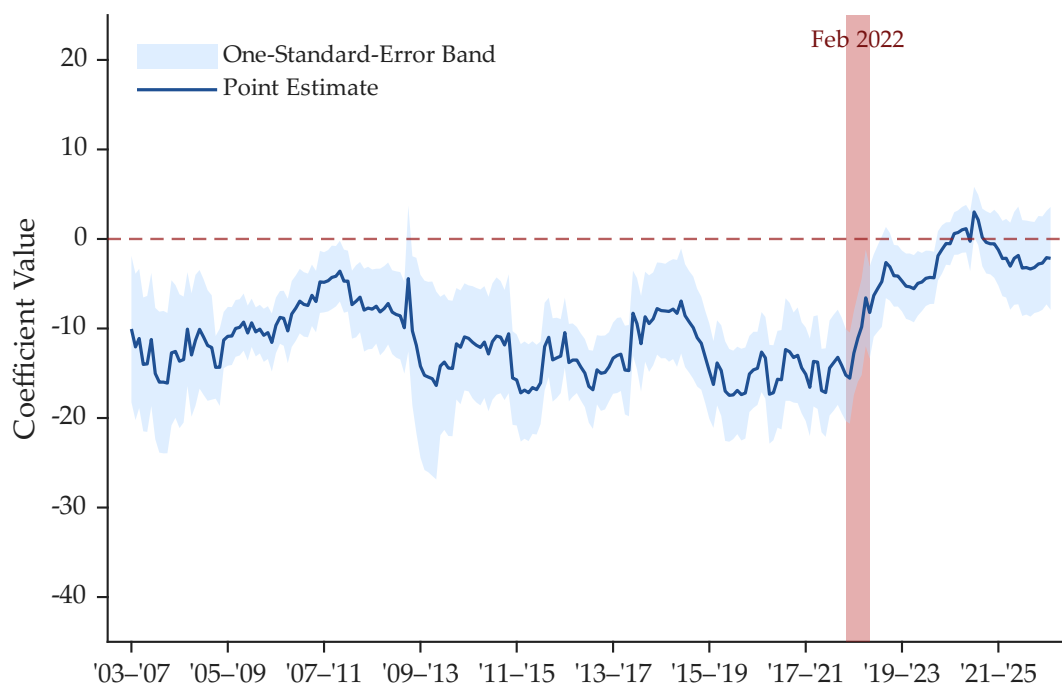


Figure 5 makes this pattern clearer. With the narrower one-standard-error band, the pre-2022 estimates are more consistently below zero, especially from the mid-2000s to the late 2010s. This suggests that the zero-inclusion problem in the 95% band is often marginal. In other words, the rolling results still support the view that the opportunity-cost channel was operating before 2022, although its strength changed over time.

After February 2022, $\hat{\beta}$ moves sharply toward zero. This shift appears in both figures. In Figure 4, the 95% confidence band includes zero throughout the post-war period. In Figure 5, the point estimate also stays much closer to zero than in most earlier windows. This suggests that gold's sensitivity to real interest rates weakened after 2022.

Overall, the rolling-window evidence supports the segmented-regression results, but with some caution. The relationship between gold returns and real interest rates appears to have weakened after 2022, but the coefficient was already time-varying before then. Therefore, we interpret the rolling evidence as suggestive rather than conclusive evidence

of structural change. The main point is not that February 2022 created a completely new relationship, but that gold's already time-varying real-rate sensitivity moved unusually close to zero after 2022 and remained there through the end of the sample.

6.5 Asymmetric Shock Response

Table 3 reports Wald test p -values for the symmetry hypothesis $H_0 : \gamma^+ = \gamma^-$ across all fifteen combinations of five safe-haven proxies and three dependent variable specifications.

Table 3: Wald Test p -Values: $H_0 : \gamma^+ = \gamma^-$

Decomposed Proxy	Dependent Variable		
	$\Delta\tilde{g}_{CPI,t}$	$\Delta g_{Oil,t}$	$\tilde{g}_{CPI,t}$
RV	0.601	0.018**	0.146
EPU	0.371	0.000***	0.000***
VIX	0.936	0.034**	0.081*
FSI	0.680	0.065*	0.712
GPR	0.787	0.509	0.821

Notes: Each cell reports the p -value from a Wald test of $H_0 : \gamma^+ = \gamma^-$ in equation (17), estimated with Newey-West standard errors (4 lags) over the full sample January 2003 – February 2026. The three dependent variables are: $\Delta\tilde{g}_{CPI,t}$ = first difference of CPI-adjusted YoY gold return ($I(1)$); $\Delta g_{Oil,t}$ = Gold-Oil ratio YoY return ($I(0)$); $\tilde{g}_{CPI,t}$ = CPI-adjusted YoY gold return in levels (for comparison). *, **, *** denote rejection of H_0 at the 10%, 5%, and 1% levels, respectively.

For $\Delta\tilde{g}_{CPI,t}$, all five tests fail to reject symmetry ($p > 0.37$), indicating no detectable ratchet effect at the monthly first-difference frequency for the main CPI-adjusted specification. The evidence is stronger for the gold-oil ratio: four out of five proxies reject symmetry, with EPU significant at the 1% level, RV and VIX significant at the 5% level, and FSI marginally significant at the 10% level. The level specification $\tilde{g}_{CPI,t}$ also rejects symmetry for EPU and marginally rejects for VIX.

These results provide suggestive, but not uniform, evidence of asymmetric responses to uncertainty shocks. The strongest evidence appears in the gold-oil ratio and in policy-uncertainty-related measures rather than in the main first-differenced CPI-adjusted specification. GPR is insignificant across all specifications, suggesting that geopolitical-risk increases and decreases do not generate a statistically detectable asymmetric response in this regression design.

Therefore, the asymmetry results should be interpreted as a possible mechanism rather than as definitive proof of a ratchet effect in all measures of gold returns. The evidence is consistent with the view that some forms of uncertainty, especially policy and financial-market uncertainty, may have more persistent effects on gold's relative value. However, the results also show that the conclusion depends on the choice of dependent variable and proxy.

7 Conclusion

This paper examines why gold prices have continued to rise since 2022 despite relatively high real interest rates. The central empirical question is whether the post-2022 surge reflects a temporary safe-haven premium or a more persistent change in gold-pricing dynamics.

Using monthly data from 2003 to 2026, we find that the traditional opportunity-cost channel was clearly visible before February 2022. In the pre-war period, increases in TIPS yields were negatively and statistically significantly associated with CPI-adjusted gold returns and gold-oil ratio returns. After February 2022, however, the negative real-rate coefficient becomes smaller in magnitude and statistically insignificant in the main first-difference specifications. Rolling-window estimates are consistent with this attenuation, as the estimated real-rate sensitivity moves toward zero after 2022.

The evidence should be interpreted cautiously. The post-2022 sample contains only 48 monthly observations, and the rolling-window coefficient is unstable in parts of the earlier sample as well. Therefore, the results do not prove that gold has permanently decoupled from real interest rates. A more defensible conclusion is that the real-rate channel has weakened since 2022 and that safe-haven demand, policy uncertainty, financial stress, and other risk-related forces appear to have become more important in explaining gold's recent behavior.

The paper also finds suggestive evidence of asymmetric responses to uncertainty shocks, especially for the gold-oil ratio and for policy-uncertainty measures. However, the ratchet-effect evidence is not uniform across all dependent variables. Future work could strengthen the analysis by reporting the full Newey-West standard errors in an appendix, conducting formal structural-break tests, and incorporating a consistent monthly measure of central-bank gold purchases or reserves if that channel remains central to the narrative.

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